Artificial Intelligence

Assignment 1

Group: Ritchie

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1. Limits of current AIs

YS how many conturies start with letter V?

As of my knowledge cutoff in September 2021, there are no countries whose names start with the letter "V." However, please note that geopolitical changes can occur over time, and new countries may emerge or change their names. It's always a good idea to consult the latest reliable sources for the most up-to-date information.

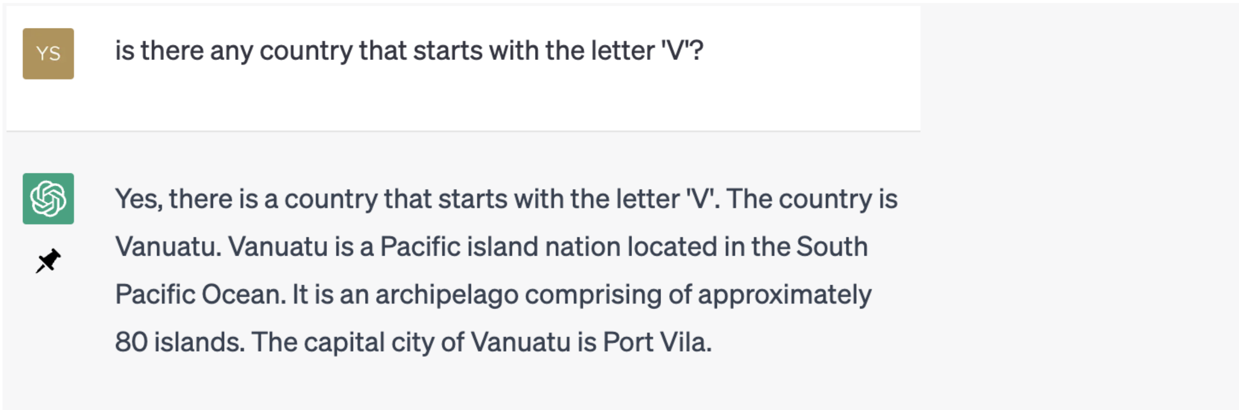
Explanation: This question idea has been gained from Reddit, a community site where people talk about interesting subjects. Chatgpt answered there are no countries, but actually there are some countries starting with letter V, such as Viet nam, Venezuela, Vatican, etc. To examine why Chatgpt answered wrong, a comparison between the answer and another similar question's answer has been conducted.

YS how many conturies start with letter D?

There are currently four countries whose names start with the letter "D":

1. Denmark
2. Djibouti
3. Dominica
4. Dominican Republic

Please note that this information is accurate as of my last knowledge update in September 2021. It's always a good idea to verify with up-to-date sources for the most current information.


YS
is there any country that starts with the letter 'V'?

Yes, there is a country that starts with the letter 'V'. The country is Vanuatu. Vanuatu is a Pacific island nation located in the South Pacific Ocean. It is an archipelago comprising of approximately 80 islands. The capital city of Vanuatu is Port Vila.

Opinion: ChatGPT is a language token generation model, which is trained on a large amount of natural language data to generate natural language token responses. Presumably, during the process of training and generating natural language, some knowledge in language may be compressed or lost. A noteworthy point is when asked about a single country starting with "V," ChatGPT can provide an answer, but when asked about how many countries start with "V," it may respond that there are none. This demonstrates that while knowledge data exists in the model, there can be loss during the process of converting that knowledge into natural language responses.

2. Modeling in propositional logic

Assume the following propositions:

A = "Alice visits the party"
B = "Bob visits the party"
C = "Charlie visits the party"
D = "Daniel visits the party"

Formulate the following sentences in propositional logic over the signature At- {A, B,C,D):

a) Alice visits the party and Bob doesn't.

$= (A \land \neg B)$

b) Daniel and Bob visit the party if and only if Charlie also visits the party.

$= (D \land B) \leftrightarrow C$

c) If Alice and Bob visit the party, then Charlie does as well-but only if Daniel is not visiting the party.

$= (A \land B) \text{->} (C \land \neg D)$

d) Charlie visits the party if and only if not both Bob and Alice visit the party, or atleast Daniel visits the party.

$= (C \leftrightarrow \neg(A \land B)) \lor D$

e) If Alice visits the party, then Bob and Charlie do as well-if Alice does not visit the party, then Charlie and Daniel visit the party.

$= (A \rightarrow (B \land C)) \land (\neg A \rightarrow (C \land D))$

3. Reasoning in propositional logic

Let At(P,Q, R) be a propositional signature. Determine which of the following formulas are valid and/or satisfiable and/or unsatisfiable.
Justify your answer, e.g. using interpretations or equivalences.

(1) $(P \land Q) \Rightarrow (P \lor Q)$

$\equiv \neg(P \land Q) \lor (P \lor Q)$
$\equiv (\neg P \lor \neg Q) \lor (P \lor Q)$
$\equiv (\neg P \lor P) \lor (\neg Q \lor Q)$
$\equiv T \lor T$
$\equiv T$

(2) $(P \lor Q) \Rightarrow (P \land Q)$

$\equiv \neg(P \lor Q) \lor (P \land Q)$
$\equiv (\neg P \land \neg Q) \lor (P \land Q)$
$\equiv (\neg P \lor P) \land (\neg P \lor Q) \land (\neg Q \lor P) \land (\neg Q \lor Q)$
$\equiv T \land (\neg P \lor Q) \land (\neg Q \lor P) \land T$
$\equiv T \land T \land (\neg P \lor Q) \land (\neg Q \lor P)$
$\equiv T \land (\neg P \lor Q) \land (\neg Q \lor P)$
$\equiv (\neg P \lor Q) \land (\neg Q \lor P)$
$\equiv P \leftrightarrow Q$

(3) $\neg(P \land \neg\neg P)$

$\equiv \neg(P) \lor \neg(\neg\neg P)$
$\equiv \neg P \lor \neg P$
$\equiv \neg(P \land P)$
$\equiv \neg P$

(4) $Q \Rightarrow \neg Q$

$\equiv \neg Q \lor \neg Q$

$\equiv \neg(Q \land Q)$

$\equiv \neg Q$

(5) $Q \land \neg Q$

$\equiv \bot$

(6) $\neg(\neg P \lor \neg\neg P)$

$\equiv \neg(\neg P) \land \neg(\neg\neg P)$

$\equiv P \land \neg P$

$\equiv \bot$

(7) $((P \Rightarrow Q) \land (\neg P \Rightarrow R)) \Rightarrow (Q \lor R)$

$\equiv ((\neg P \lor Q) \land (\neg (\neg P) \lor R)) \Rightarrow (Q \lor R)$

$\equiv ((\neg P \lor Q) \land (P \lor R)) \Rightarrow (Q \lor R)$

$\equiv ((\neg P \lor Q) \land P) \lor ((\neg P \lor Q) \land R) \Rightarrow (Q \lor R)$

$\equiv ((\neg P \land P) \lor (Q \land P) \lor (\neg P \land R) \lor (Q \land R)) \Rightarrow (Q \lor R)$

$\equiv (\bot \lor (Q \land P) \lor (\neg P \land R) \lor (Q \land R)) \Rightarrow (Q \lor R)$

$\equiv ((Q \land P) \lor (\neg P \land R) \lor (Q \land R)) \Rightarrow (Q \lor R)$

$\equiv \neg((Q \land P) \lor (\neg P \land R) \lor (Q \land R)) \lor (Q \lor R)$

$\equiv \neg((Q \land P) \lor (\neg P \land R)) \land \neg(Q \land R) \lor (Q \lor R)$

$\equiv \neg((Q \land P) \lor (\neg P \land R)) \land T$

$\equiv (\neg Q \lor \neg P) \land (P \lor \neg R) \land T$